A SUM–RECIPROCAL INEQUALITY AND TWO APPLICATIONS

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Abstract. We give a short derivation of the inequality $(x_1 + \cdots + x_n)(1/x_1 + \cdots + 1/x_n) \geq n^2$ for positive reals from the Cauchy–Schwarz inequality, deduce the comparison of the arithmetic and harmonic means and Nesbitt's inequality, and record an elementary bound on $(1+1/n)^n$ obtained from Bernoulli's inequality.

1. Introduction

The inequality in Theorem 2.6 below is a standard first application of the Cauchy–Schwarz inequality [1]. This note is intended as a self-contained reference for it and for a few consequences we use elsewhere. Section 2 contains the main theorem and its corollaries, including Nesbitt's inequality [3]; Section 3 treats the sequence $(1 + 1/n)^n$.

2. The sum–reciprocal inequality

Throughout this section $n \geq 1$ is an integer and $x = (x_1, \ldots, x_n)$ is an n-tuple of real numbers; all sums run over $i = 1, \ldots, n$.

Definition 2.1. Let x be an n-tuple of positive reals. Its *arithmetic mean* and its *harmonic mean* are

$$A(x) = \frac{1}{n}\sum x_i, \qquad H(x) = \frac{1}{n}\sum \frac{1}{x_i}.$$

Lemma 2.2. *For all real a, b we have $a^2 + b^2 \geq 2ab$.*

Proof. $a^2 + b^2 - 2ab = (a - b)^2 \geq 0$. □

Lemma 2.3. *For every real $x \neq 0$ we have $x + \frac{1}{x} \geq 2$.*

Proof. Multiplying by x, the claim becomes $x^2 + 1 \geq 2x$, which is Lemma 2.2 with $a = x$, $b = 1$. □

Lemma 2.4 (Cauchy–Schwarz [1])**.** *For real n-tuples a, b we have $\left(\sum a_i b_i\right)^2 \leq \left(\sum a_i^2\right)\left(\sum b_i^2\right)$.*

Proof. If all $a_i = 0$ both sides vanish. Otherwise the quadratic polynomial $q(t) = \sum (a_i t + b_i)^2 = t^2 \sum a_i^2 + 2t \sum a_i b_i + \sum b_i^2$ is non-negative for all real t and has positive leading coefficient, so its discriminant $4(\sum a_i b_i)^2 - 4 \sum a_i^2 \sum b_i^2$ is at most 0. □

Lemma 2.5. *Let $x_1, \dots, x_n$ be real numbers with $x_i \geq 0$ for all i. Then $\left(\sum x_i\right)^2 \leq n \sum x_i^2$.*

Proof. Apply Lemma 2.4 with $a_i = x_i$ and $b_i = x_i$ for all i, and use $\sum 1^2 = n$. □

Theorem 2.6 (Main theorem)**.** *Let $n \geq 1$ and let $x_1, \dots, x_n$ be positive real numbers. Then*

$$\Big(\sum x_i\Big)\Big(\sum \frac{1}{x_i}\Big) \geq n^2.$$

Proof. Apply Lemma 2.4 with $a_i = \sqrt{x_i}$ and $b_i = 1/\sqrt{x_i}$. Then $a_i b_i = 1$, so the left-hand side of Lemma 2.4 is n^2, while the right-hand side is $\big(\sum x_i\big)\big(\sum 1/x_i\big)$. □

Corollary 2.7 (AM–HM)**.** *For every n-tuple x of positive reals, $A(x) \geq H(x)$.*

Proof. Dividing Theorem 2.6 by $n \sum \frac{1}{x_i}$ gives $\frac{1}{n}\sum x_i \geq n \big/ \sum \frac{1}{x_i}$, which is the claim. □

Corollary 2.8. *Let $x_1, \dots, x_n$ be positive reals with $\sum x_i = 1$ and $\sum 1/x_i < n^2$. Then the x_i are not all equal.*

Proof. If $x_i = 1/n$ for all i, then $\sum 1/x_i = n^2$, contrary to the hypothesis. □

Proposition 2.9 (Nesbitt [3])**.** *For all positive reals a, b, c,*

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Proof. Put $s = a+b+c$ and apply Theorem 2.6 with $n = 3$ and $(x_1, x_2, x_3) = (b+c,\ c+a,\ a+b)$, whose sum is $2s$. This gives $\sum \frac{1}{b+c} \geq \frac{9}{2s}$. Since $\frac{a}{b+c} = \frac{s}{b+c} - 1$, the left-hand side equals $s \sum \frac{1}{b+c} - 3 \geq \frac{9}{2} - 3 = \frac{3}{2}$. □

3. Bernoulli's inequality and $(1 + 1/n)^n$

We use Bernoulli's inequality in the form found in [2]: *for every real x and every $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1 + nx.$$

For $n = 0$ both sides equal 1, and the general case follows by induction on n.

Proposition 3.1. *For every $n \in \mathbb{N}$ we have $\left(1 + \frac{1}{n}\right)^n \geq 2$.*

Proof. Apply Bernoulli's inequality with $x = 1/n$: $\left(1 + \frac{1}{n}\right)^n \geq 1 + n \cdot \frac{1}{n} = 2$. □

Remark. Proposition 3.1 together with Lemma 2.3 shows that $(1+\frac{1}{n})^n + (1+\frac{1}{n})^{-n} \geq 2$ as well, but this weaker statement already follows from Lemma 2.3 alone.

References

[1] A.-L. Cauchy, *Cours d'analyse de l'École Royale Polytechnique*, Paris, 1821.
[2] G. H. Hardy, J. E. Littlewood and G. Pólya, *Inequalities*, Cambridge University Press, 1934.
[3] A. M. Nesbitt, Problem 15114, *Educational Times* **3** (1903), 37–38.

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